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Implementing Hypervisor-Specific Mitigations for Microarchitectural Data Sampling (MDS) Vulnerabilities (CVE-2018-12126, CVE-2018-12127, CVE-2018-12130, and CVE-2019-11091) in vSphere (67577)

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Purpose

This article documents the Hypervisor-Specific Mitigations enablement process required to address Microarchitectural Data Sampling (MDS) Vulnerabilities identified by CVE-2018-12126, CVE-2018-12127, CVE-2018-12130, and CVE-2019-11091 in vSphere.

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In addition to the Hypervisor-Specific Mitigations described in this article, Hypervisor-Assisted Guest Mitigations and Operating System-Specific Mitigations are also required.

These additional mitigations are documented VMSA-2019-0008.

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Introduction to MDS

Intel has disclosed details on a new wave of speculative-execution vulnerabilities known collectively as “Microarchitectural Data Sampling (MDS)" that can occur on Intel microarchitecture prior to 2nd Generation Intel® Xeon® Scalable Processors (formerly known as Cascade Lake). These issues may allow a malicious user who can locally execute code on a system to infer the values of data otherwise protected by architectural mechanisms.

Attack Vector Summary

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- Sequential-context attack vector (Inter-VM): a malicious VM can potentially infer recently accessed data of a previous context (hypervisor thread or other VM thread) on either logical processor of a processor core.

- Concurrent-context attack vector (Inter-VM): a malicious VM can potentially infer recently accessed data of a concurrently executing context (hypervisor thread or other VM thread) on the other logical processor of the Hyper-Threading-enabled processor core.

Mitigation Summary

- The Sequential-context attack vector (Inter-VM): is mitigated by a Hypervisor update to the product versions listed in VMSA-2019-0008 section 3a. These mitigations are dependent on Intel microcode updates (provided in separate ESXi patches for most Intel hardware platforms) also listed in the advisory. This mitigation is enabled by default and does not impose a significant performance impact.

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- The Concurrent-context attack vector (Inter-VM): is mitigated through enablement of the ESXi Side-Channel-Aware Scheduler Version 1 or Version 2. These options may

impose a non-trivial performance impact and are not enabled by default.

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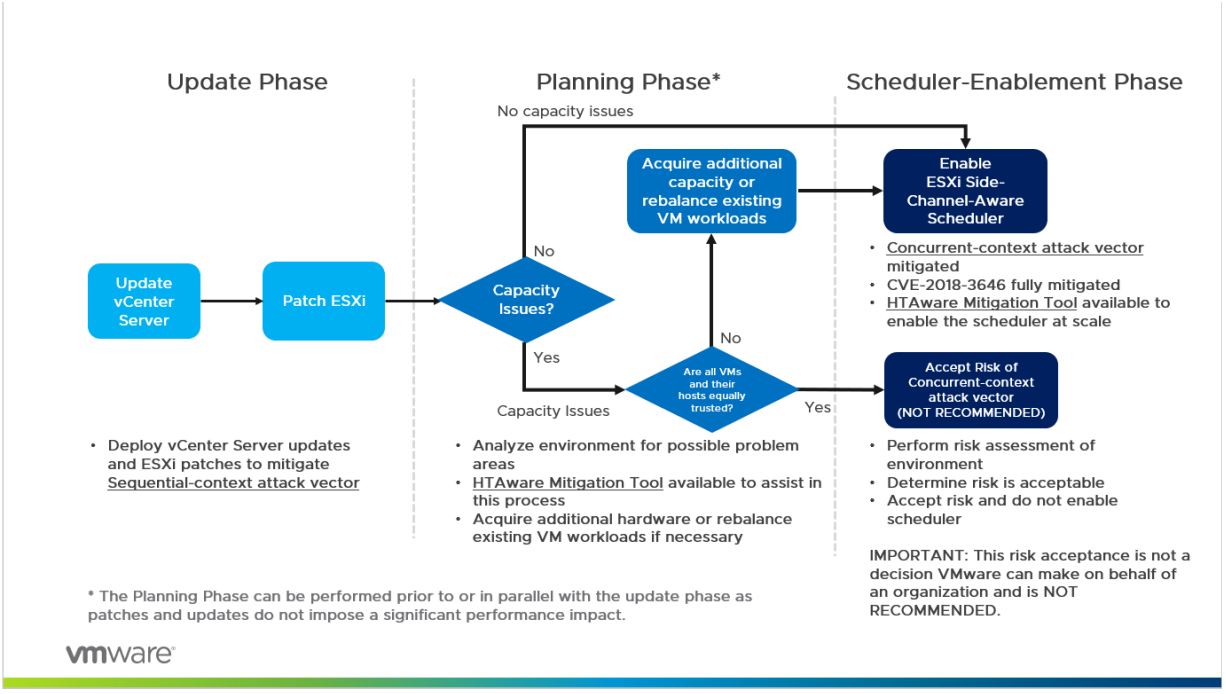
Important: Disabling Intel Hyperthreading in firmware/BIOS (or by using VMkernel.Boot.Hyperthreading) after applying vSphere updates and patches is not recommended.

Resolution

The Inter-VM mitigation process for MDS is divided into three phases:



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1. **Update Phase: Apply vSphere Updates and Patches**

   The Inter-VM Sequential-context attack vector is mitigated by a vSphere update to the product versions listed in VMware Security Advisory VMSA-2019-0008. This mitigation is dependent on Intel microcode updates (provided in separate ESXi patches for most Intel hardware platforms) which are also documented in VMSA-2019-0008. This mitigation is enabled by default and does not impose a significant performance impact.

   Note: As displayed in the workflow above, vCenter Server should be updated prior to applying ESXi patches. Notification messages were previously added in the patches to explain that the ESXi Side-Channel-Aware Scheduler must be enabled to mitigate Inter-VM Concurrent-context attack vectors. If ESXi is updated prior to vCenter you may receive cryptic notification messages relating to this. After vCenter has been

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updated, the notifications will be shown correctly.

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2. **Planning Phase: Assess Your Environment**

Inter-VM Concurrent-context attack vectors are mitigated through enablement of the ESXi Side-Channel-Aware Scheduler which is included in the updates and patches listed in VMSA-2019-0008. This scheduler is not enabled by default. Enablement of this scheduler may impose a non-trivial performance impact on applications running in a vSphere environment. The goal of the Planning Phase is to understand if your current environment has sufficient CPU capacity to enable the scheduler without operational impact.

The following list summarizes potential problem areas after enabling the ESXi Side-Channel-Aware Scheduler:

- VMs configured with vCPUs greater than the physical cores available on the ESXi host
- VMs configured with custom affinity or NUMA settings
- VMs with latency-sensitive configuration
- ESXi hosts with Average CPU Usage greater than 70%
- Hosts with custom CPU resource management options enabled
- HA Clusters where a rolling upgrade will increase Average CPU Usage above 100%

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Important: The above list is meant to be a brief overview of potential problem areas related to enablement of the ESXi Side-Channel-Aware Scheduler. The VMware Performance Team has provided an in-depth guide as well as performance data in KB55767 (/s/article/55767). It is strongly suggested to thoroughly review this document prior to enablement of the scheduler.

Note: It may be necessary to acquire additional hardware, or rebalance existing workloads, before enablement of the ESXi Side-Channel-Aware Scheduler. Organizations can choose not to enable the ESXi Side-Channel-Aware Scheduler after by accepting the risk posed by the Concurrent-context attack vector. This is NOT RECOMMENDED and VMware cannot make this decision on behalf of an organization.

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3. **Scheduler-Enablement Phase:**
   1. **Enable the ESXi Side-Channel-Aware Scheduler in ESXi 5.5, 6.0, 6.5, and 6.7 prior to 6.7u2.**

After addressing the potential problem areas described above during the Planning Phase, the ESXi Side-Channel-Aware Scheduler must be enabled to mitigate Inter-VM Concurrent-context attack vectors. The scheduler can be enabled on an individual ESXi host via the advanced configuration option hyperthreadingMitigation.

Note: Enabling this option will result in the vSphere UI reporting only a single logical processor per physical core; halving the number of logical processors if Hyperthreading was previously enabled. In addition Hyperthreading may be reported as 'Disabled' in various configuration tabs.

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Enabling the ESXi Side-Channel-Aware Scheduler using the vSphere Web Client or vSphere Client

1. Connect to the vCenter Server using either the vSphere Web or vSphere Client.
2. Select an ESXi host in the inventory.
3. Click the **Manage** (5.5/6.0) or **Configure** (6.5/6.7) tab.
4. Click the **Settings** sub-tab (5.5/6.0) or find the **System** section in the middle panel (6.5/6.7).
5. Under the System heading, click **Advanced System Settings**.
6. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigation`
7. Select the setting by name and click the **Edit** pencil icon.
8. Change the configuration option to true (default: false).
9. Click **OK**.
10. Reboot the ESXi host for the configuration change to go into effect.

Enabling the ESXi Side-Channel-Aware Scheduler using ESXi Embedded Host Client

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1. Connect to the ESXi host by opening a web browser to

`https://HOSTNAME`.

2. Click the **Manage** tab.
3. Click the **Advanced settings** sub-tab
4. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigation`
5. Select the setting by name and click the **Edit** pencil icon
6. Change the configuration option to true (default: false)
7. Click **Save**.
8. Reboot the ESXi host for the configuration change to go into effect.

Enable ESXi Side-Channel-Aware Scheduler setting using ESXCLI

1. SSH to an ESXi host or open a console where the remote ESXCLI is installed. For more information, see the http://www.vmware.com/support/developer/vcli/ (http://www.vmware.com/support/developer/vcli/)
2. Check the current runtime value of the HTAware Mitigation Setting by running `esxcli system settings kernel list -o hyperthreadingMitigation`
3. To enable HT Aware Mitigation, run this command:

```
esxcli system settings kernel set -s
hyperthreadingMitigation -v TRUE
```

4. Reboot the ESXi host for the configuration change to go into effect.

2. **Enable the ESXi Side-Channel-Aware Scheduler (SCAv1) or the ESXi Side-Channel-Aware Scheduler v2 (SCAv2) in ESXi 6.7u2 (13006603) or later**

Note: ESXi 6.7u2 (13006603) and future release lines of ESXi include the ESXi Side-Channel-Aware Scheduler v2. Prior release lines such as 6.5, 6.0, and 5.5 cannot accommodate this new scheduler.

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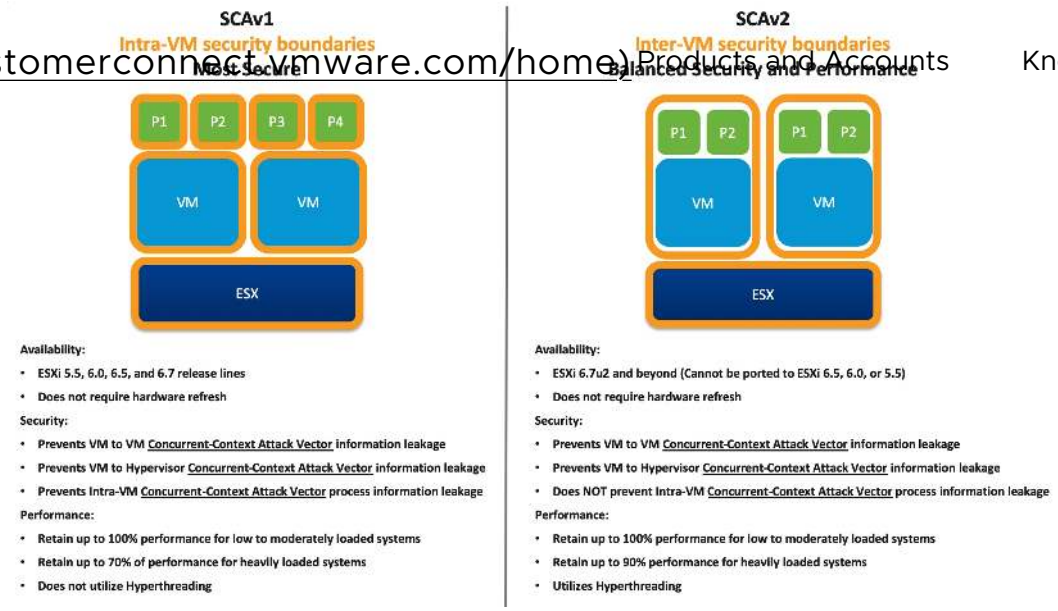

VMware has published a white paper entitled Performance of vSphere 6.7 Scheduling Options (https://www.vmware.com/techpapers/2018/scheduler-options-vsphere67u2-perf.html) which provides a more detailed look into the performance differences between SCAv1 and SCAv2. Please review this document before continuing.

Enabling the ESXi Side-Channel-Aware Scheduler (SCAv1) using the vSphere Web Client or vSphere Client

1. Connect to the vCenter Server using either the vSphere Web or vSphere Client.
2. Select an ESXi host in the inventory.
3. Click the **Configure** tab.
4. Under the System heading, click **Advanced System Settings**.
5. Click **Edit**.
6. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigation`
7. Select the setting by name
8. Change the configuration option to true (default: false).
9. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigationIntraVM`
10. Change the configuration option to true (default: true).
11. Click **OK**.
12. Reboot the ESXi host for the configuration change to go into effect.

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Enabling the ESXi Side-Channel-Aware Scheduler (SCAv1) using ESXi Embedded Host Client

1. Connect to the ESXi host by opening a web browser to `https://HOSTNAME`.
2. Click **Manage** under host navigator
3. Click the **Advanced settings** Tab
4. Use the search box to find `VMkernel.Boot.hyperthreadingMitigation`
5. Select the `VMkernel.Boot.hyperthreadingMitigation` setting and click the **Edit** Option
6. Change the configuration option to true (default: false)
7. Click **Save**.
8. Use the search box to find `VMkernel.Boot.hyperthreadingMitigationIntraVM`
9. Select the `VMkernel.Boot.hyperthreadingMitigationIntraVM` setting and click the **Edit** Option
10. Change the configuration option to true (default: true).
11. Click **Save**.
12. Reboot the ESXi host for the configuration change to go into effect.

Enable ESXi Side-Channel-Aware Scheduler (SCAv1) using ESXCLI

1. SSH to an ESXi host or open a console where the remote ESXCLI is installed. For more information, see the http://www.vmware.com/support/developer/vcli/ (http://www.vmware.com/support/developer/vcli/).
2. Check the current runtime values by running `esxcli system settings kernel list -o hyperthreadingMitigation and esxcli system settings kernel list -o hyperthreadingMitigationIntraVM`
3. To enable the ESXi Side-Channel-Aware Scheduler Version 1, run these commands:

```
esxcli system settings kernel set -s
hyperthreadingMitigation -v TRUE
```

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```
esxcli system settings kernel set -s
hyperthreadingMitigationIntraVM -v TRUE
```

4. Reboot the ESXi host for the configuration change to go into effect.

Enabling the ESXi Side-Channel-Aware Scheduler Version 2 (SCAv2) using the vSphere Web Client or vSphere Client

1. Connect to the vCenter Server using either the vSphere Web or vSphere Client.
2. Select an ESXi host in the inventory.
3. Click the **Configure** tab.
4. Under the System heading, click **Advanced System Settings**.
5. Click **Edit**.
6. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigation`
7. Select the setting by name
8. Change the configuration option to true (default: false).
9. Click in the Filter box and search for `VMkernel.Boot.hyperthreadingMitigationIntraVM`
10. Change the configuration option to false (default: true).
11. Click **OK**.
12. Reboot the ESXi host for the configuration change to go into effect.

Enabling the ESXi Side-Channel-Aware Scheduler Version 2 (SCAv2) using ESXi Embedded Host Client

1. Connect to the ESXi host by opening a web browser to `https://HOSTNAME`.
2. Click **Manage** under host navigator
3. Click the **Advanced Settings** tab
4. Use the search box to find `VMkernel.Boot.hyperthreadingMitigation`
5. Select the `VMkernel.Boot.hyperthreadingMitigation` setting and click the **Edit** option
6. Change the configuration option to true (default: false)
7. Click **Save**.
8. Use the search box to find `VMkernel.Boot.hyperthreadingMitigationIntraVM`

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9. Select the `VMkernel.Boot.hyperthreadingMitigationIntraVM` setting and click the **Edit** option.
10. Change the configuration option to false (default: true).
11. Click **Save**.
12. Reboot the ESXi host for the configuration change to go into effect.

Enable ESXi Side-Channel-Aware Scheduler Version 2 (SCAv2) using ESXCLI

1. SSH to an ESXi host or open a console where the remote ESXCLI is installed. For more information, see the http://www.vmware.com/support/developer/vcli/ (http://www.vmware.com/support/developer/vcli/)
2. Check the current runtime values by running `esxcli system settings kernel list -o hyperthreadingMitigation` and `esxcli system settings kernel list -o hyperthreadingMitigationIntraVM`
3. To enable the ESXi Side-Channel-Aware Scheduler Version 2 run these commands:

```
esxcli system settings kernel set -s
hyperthreadingMitigation -v TRUE
```

```
esxcli system settings kernel set -s
hyperthreadingMitigationIntraVM -v FALSE
```

4. Reboot the ESXi host for the configuration change to go into effect.

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ESXi 6.7u2 (and later) Scheduler Configuration Summary

hyperthreadingMitigation	hyperthreadingMitigationIntraVM	Scheduler Enabled
FALSE	TRUE or FALSE	Default scheduler (unmitigated)
TRUE	TRUE	SCAv1

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TRUE	FALSE	SCAv2

HTAware Mitigation Tool

VMware has provided a tool to assist in performing both the **Planning Phase** and the **Scheduler-Enablement** Phase at scale. This tool has been updated to include SCAv2 support and can be found in KB56931 (/s/article/56931) along with detailed instructions on its usage, capabilities, and limitations.

Table 1: Affected Intel Processors Supported by ESXi

Intel Code Name	FMS	Intel Brand Names
Nehalem-EP	0x106a5	Intel Xeon 35xx Series; Intel Xeon 55xx Series
Lynnfield	0x106e5	Intel Xeon 34xx Lynnfield Series
Clarkdale	0x20652	Intel i3/i5 Clarkdale Series; Intel Xeon 34xx Clarkdale Series
Arrandale	0x20655	Intel Core i7-620LE Processor
Sandy Bridge DT	0x206a7	Intel Xeon E3-1100 Series; Intel Xeon E3-1200 Series; Intel i7-2655-LE Series; Intel i3-2100 Series
Westmere EP	0x206c2	Intel Xeon 56xx Series; Intel Xeon 36xx Series
Sandy Bridge EP	0x206d7	Intel Pentium 1400 Series; Intel Xeon E5-1400 Series; Intel Xeon E5-1600 Series; Intel Xeon E5-2400 Series; Intel Xeon E5-2600 Series; Intel Xeon E5-4600 Series
Nehalem EX	0x206e6	Intel Xeon 65xx Series; Intel Xeon 75xx Series

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Westmere EX	0x206f2	Intel Xeon E7-8800 Series; Intel Xeon E7-4800 Series; Intel Xeon E7-2800 Series
Ivy Bridge DT	0x306a9	Intel i3-3200 Series; Intel i7-3500-LE/UE, Intel i7-3600-QE, Intel Xeon E3-1200-v2 Series; Intel Xeon E3-1100-C-v2 Series; Intel Pentium B925C
Haswell DT	0x306c3	Intel Xeon E3-1200-v3 Series
Ivy Bridge EP	0x306e4	Intel Xeon E5-4600-v2 Series; Intel Xeon E5-2400-v2 Series; Intel Xeon E5-2600-v2 Series; Intel Xeon E5-1400-v2 Series; Intel Xeon E5-2600-v2 Series
Ivy Bridge EX	0x306e7	Intel Xeon E7-8800/4800/2800-v2 Series
Haswell EP	0x306f2	Intel Xeon E5-2400-v3 Series; Intel Xeon E5-1400-v3 Series; Intel Xeon E5-1600-v3 Series; Intel Xeon E5-2600-v3 Series; Intel Xeon E5-4600-v3 Series
Haswell EX	0x306f4	Intel Xeon E7-8800/4800-v3 Series
Broadwell H	0x40671	Intel Core i7-5700EQ; Intel Xeon E3-1200-v4 Series
Avoton	0x406d8	Intel Atom C2300 Series; Intel Atom C2500 Series; Intel Atom C2700 Series
Broadwell EP/EX	0x406f1	Intel Xeon E7-8800/4800-v4 Series; Intel Xeon E5-4600-v4 Series; Intel Xeon E5-2600-v4 Series; Intel Xeon E5-1600-v4 Series
Skylake SP	0x50654	Intel Xeon Platinum 8100 (Skylake-SP) Series; Intel Xeon Gold 6100/5100 (Skylake-SP) Series Intel Xeon Silver 4100, Bronze 3100 (Skylake-SP) Series

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Broadwell DE	0x50662	Intel Xeon D-1500 Series
Broadwell DE	0x50663	Intel Xeon D-1500 Series
Broadwell DE	0x50664	Intel Xeon D-1500 Series
Broadwell NS	0x50665	Intel Xeon D-1500 Series
Skylake H/S	0x506e3	Intel Xeon E3-1500-v5 Series; Intel Xeon E3-1200-v5 Series
Kaby Lake H/S/X	0x906e9	Intel Xeon E3-1200-v6

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